

Abhiroop Sikand

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EDUCATION

Bachelor of Engineering, *Specializing in Software Engineering – Big Data*

2022 - 2026

York University - Academic Excellence Scholarship (95%+)

- Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Systems, Software Design, Computer Organization, Software Tools, Electricity

WORK EXPERIENCE

Web Developer:

May 2025 - Present

Royal Canadian Realty

- Developed and deployed a responsive **landing page**, improving lead generation and online presence by **23%**.
- Used **HTML/CSS**, **JavaScript**, and **React** to build an engaging design showcasing listings and contact features.
- Collaborated directly with stakeholders to tailor branding, content, and layout for client needs.

Coding Tutor:

September 2024 – Present

Middlefield Collegiate

- Mentored students in Java programming, covering fundamental concepts, **OOP principles**, and **data structures**.
- Provided personalized guidance and **debugging** assistance, improving coding proficiency and problem-solving.

PROJECTS

YorkU AI Chatbot – Virtual Campus Assistant:

- Designed and deployed an intelligent chatbot using **Python** and OpenAI's GPT **API**, tailored to answer frequently asked questions about York University.
- Embedded **prompt engineering** to guide AI responses and maintain relevance to YorkU services and terminology.
- Achieved high accuracy in responding to over **50 curated queries** by students, demonstrating practical **AI integration** and **usability**.

Restaurant Reviewer App:

- Led a team** to design an end-to-end application using **Java**, projected to increase restaurant visibility by **45%**.
- Developed and implemented an optimized **SQL** database architecture with **38% faster response times**.
- Created effective user interface illustrating proficiency in front-end design.

Chrome Extension – Article Summarizer:

- Designed a software tool using JavaScript and HTML/CSS to summarize articles, reducing reading time by **60-70%**.
- Integrated API to quickly extract key information from lengthy articles, enhancing efficiency and comprehension.

Yorku Parking System:

- Built a parking management system in Java with a JavaFX UI, managing **500+ simulated parking spots**.
- Implemented key features such as real-time availability, and booking, improving system usability by **70%**.
- Developed user and role management (students, faculty, managers etc.) with auto-generated admin accounts, parking validation, rate differentiation, and sensor-based spot tracking.

Autonomous Vehicle:

- Created a robotic system using **C++** featuring path following and obstacle avoidance with **96% accuracy**.
- Programmed microcontroller and ultrasonic/IR sensors to optimize the embedded systems algorithms.

Space Invaders:

- Using expertise in **software/game development** to successfully build a Space Invaders-style game in Java with comprehensive user interface features and high-score functionality.

TECHNICAL SKILLS

Programming Languages: Java, Python, C, C++, C#, HTML/CSS, MATLAB, JavaScript, SQL, RISC-V, Verilog, Event-B

Tools & Frameworks: Git, React, Spring Boot, AWS, Agile, CMD, Maven, DevOps, Quartus, Microsoft Suit, Google Workspace

LEADERSHIP EXPERIENCE

Charity - Crafting for A Cure: Vice President

September 2023 - Present

- Organized hospital and event planning efforts for charitable initiatives, ensuring impactful outcomes.